

Use Quicksort to sort the array

(1) First element as pivot

Arr = [157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 183]

Step 1

151 as pivot

↑ low
[157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 183]
↓ high

P
i++ continues as long as the element > 151.
j-- continues as long as this element < 151.

Step 2

↑ L × ✓
[183, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 157]
↓ ↑
P L

Swap (147, 117)

↑ new L
[183, 110, 117, 122, 111, 149, 151, 141, 123, 112, 147, 157]
↓ ↓ new h
P L++ h--

Step 3

P L H
↑ ↑ ↑
[183, 110, 117, 122, 111, 149, 151, 141, 123, 112, 147, 157]

Swap (149, 112)

↓

[183, 110, 117, 122, 111, 112, 149, 151, 141, 123, 147, 157]
↓ ↓
P H

Step 3

↓ newL ↓ newH
[133, 110, 117, 122, 111, 149, 151, 141, 123, 142, 147, 157]

swap(149, 112)

↓ newL ↓ newH
[133, 110, 117, 122, 111, 122, 151, 141, 123, 149, 147, 157]

↓
pivot

Step 4

↓ ↓
[133, 110, 117, 122, 111, 142, 151, 141, 123, 149, 147, 157]

↓
P

swap [151, 123]

↓ ↓
[133, 110, 117, 122, 111, 112, 141, 151, 149, 147, 157]
↑ ↑
H H

∴ since $(i < j)$, swap pivot with ↓

Step 5

↓ newL ↓ newH ↓ P ↓ newL
[133, 110, 117, 122, 111, 112, 133, 141, 151, 149, 147, 157]
↓ ↓ ↓
newL newH newL

~~swap(117, 111)~~, ~~NO swap~~

✗

~~[112, 110, 111, 122, 11]~~

swap(123, 112), NO swap

↓ newL ↓ newH ↓ P ↓ newL
[112, 110, 117, 122, 111, 123, 133, 141, 151, 149, 147, 157]
↓ ↓ ↓
newL newH newL

Step 6

Swap (117, 111) , swap (151, 147)

↓ ↓
[112, 110, 111, 122, 117, 123, 133, 141, 147, 149, 151, 157]
↑ ↑
new L H

∴ since (L) swap pivot with j

Step 7

↓ ↓ ↓ ↓
[111, 110, 112, 122, 117, 123, 133, 141, 147, 149, 151, 157]
↑ ↑ ↑ ↑
L H L H

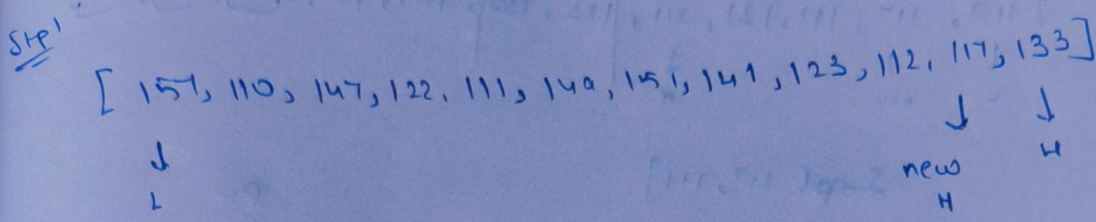
Swap (111, 110) swap (122, 117)

Step 8

[110, 111, 112, 117, 122, 123, 133, 141, 147, 149, 151, 157]

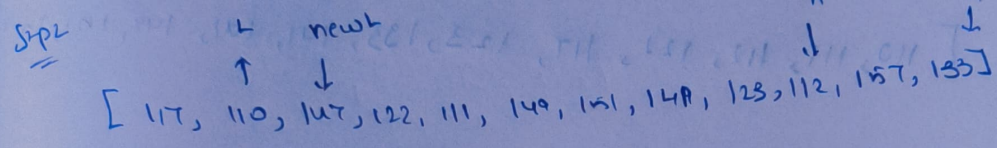
∴ Sorted. //

(11) Last element as pivot

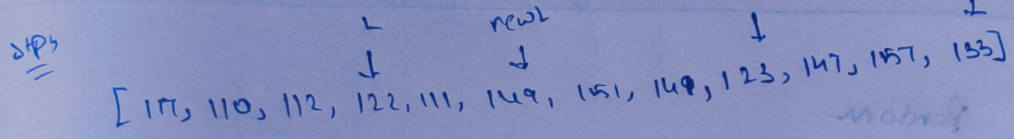


~~Since (L+1) ! > 157~~
sorted at last.

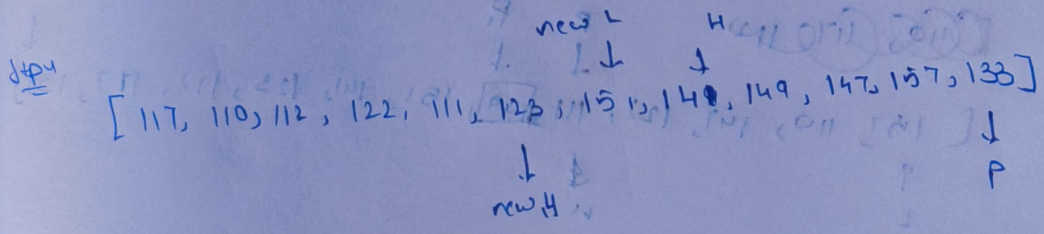
Swap (157, 117)



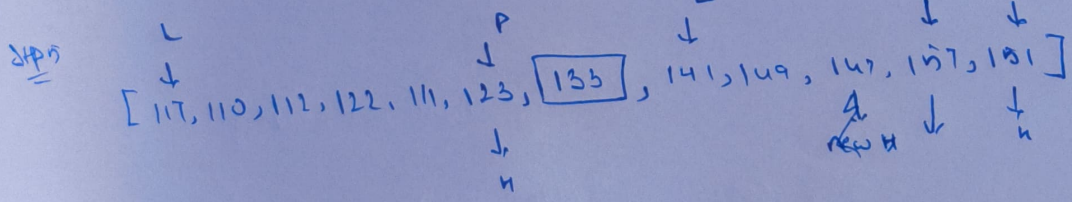
Swap (147, 112)



Swap (149, 123)

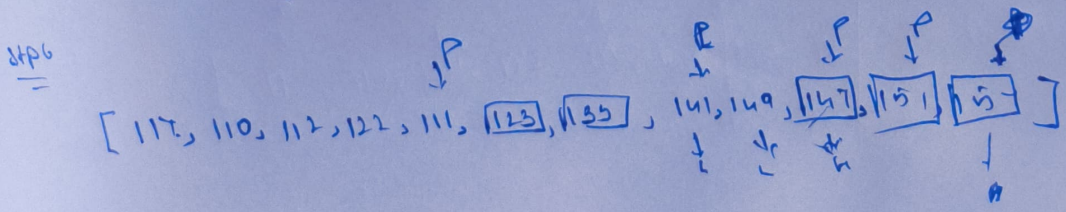


Swap (pivot, j)

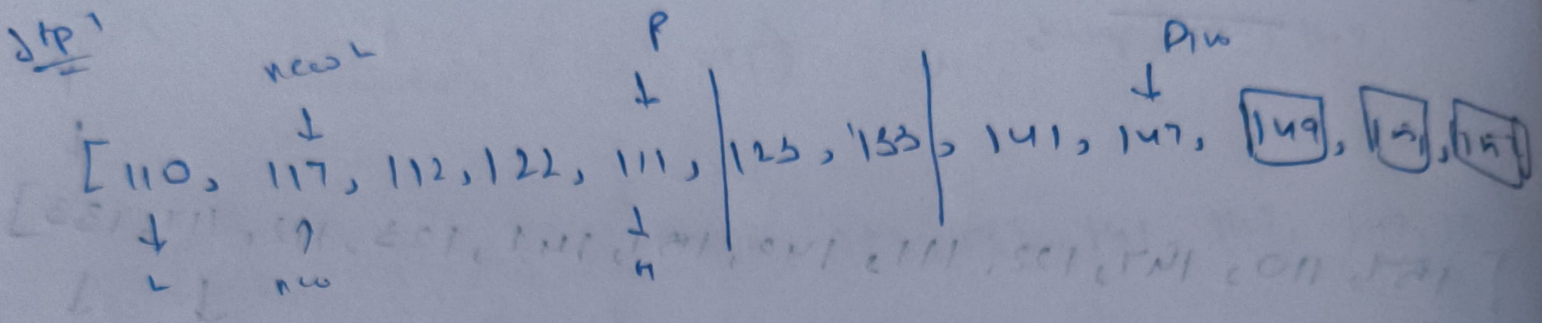


(No swap)

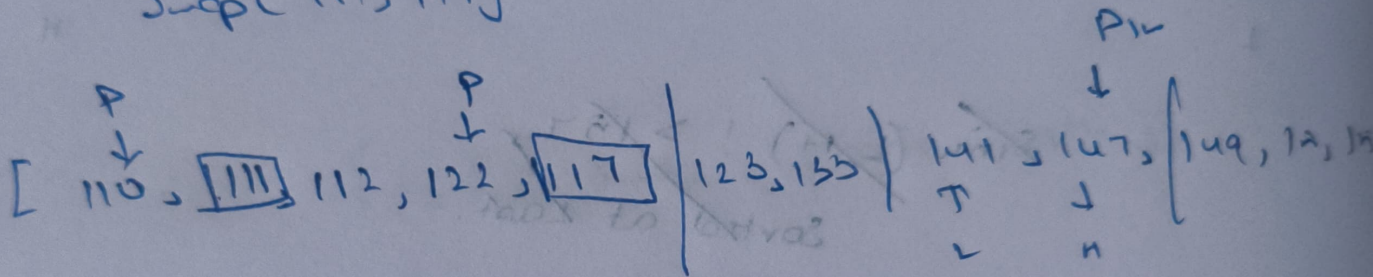
Swap (pivot, 157)



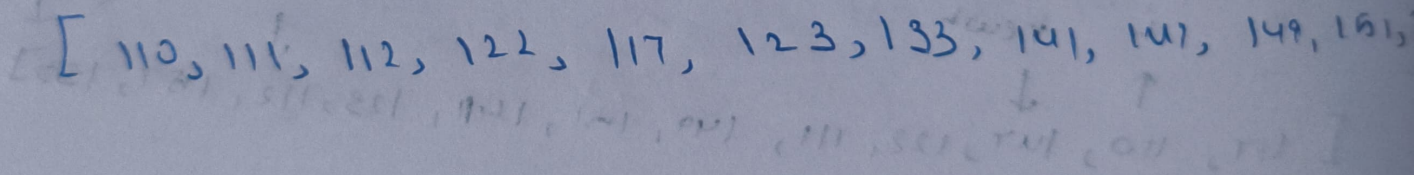
Step 1



Swap(117, 111)



Step 8



Sorted